Power and Sample Size Calculations

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Overview

- Sample size and power calculations
- Observational studies vs. RCTs
- Some software tools and additional resources
- Sample size and power for cluster RCTs
- Simulation-based approaches
- Supplemental
  - Some algebra: rearranging the sample size formula to get power
  - Sample size and power for interactions
  - Sample size and power for mediation

[0] Acknowledgments to Tyler VanderWeele for previous versions of these slides

Why sample size and power calculations

- Sample size calculations are an essential part of planning studies
- Sample size requirements should be considered early in the planning phase of a study
- A well-designed trial will be large enough to detect clinically important differences between groups with high probability
- To perform sample size calculations, we need well defined study endpoints, hypotheses, and statistical tests
- We also need estimates of the variability of the outcomes
- Both analytically and empirically coming up with estimates of the variance is often the most challenging step

Sample size

Suppose we have a continuous outcome Y and wish to test the null hypothesis

$$H_0 : \mathbb{E}[Y \mid A = 1] = \mathbb{E}[Y \mid A = 0] \qquad \text{(i.e. no treatment effect)}$$

versus the alternative hypothesis

$$H_A : \mathbb{E}[Y \mid A = 1] \neq \mathbb{E}[Y \mid A = 0] \qquad \text{(i.e. non-zero treatment effect)}$$

in a randomized trial with equally-sized treatment groups.

What is the required sample size that we need to test this hypothesis?

We need to specify:

- The significance level (Type I error rate): α (e.g. 5%)
- Power (1 - Type II error rate): $P = 1 - \beta$ (e.g. $P = 80\%$)
- The magnitude of the effect: $\eta = \mathbb{E}[Y \mid A = 1] - \mathbb{E}[Y \mid A = 0]$ that we wish to be able to detect

Sample size

To calculate the sample size **for each group** required to detect a treatment effect of size η using a two-sample t-test with significance level α and power $P = 1 - \beta$), we can use

$$N = \frac{(Z_{1-\alpha/2} + Z_P)^2 \times V}{\eta^2}$$

where $Z_{1-\alpha/2}$ and Z_P denote the $(1 - \alpha/2)$th and Pth quantiles of a standard normal variable, e.g. $Z_{1-\alpha/2} = 1.96$ for $\alpha = 0.05$ and $_P = 0.84$ for $P = 0.80$.

And where if the variance of Y is assumed to be the same in the treatment and control groups, then $V = 2\sigma^2$ where σ^2 is the variance of Y.

(Other formulas are available if the variance is different across groups or if treatment and control groups are difference sizes).

Example

Suppose we are testing a drug to lower LDL cholesterol and want to know the sample size required to detect an effect size of 13 point decline with power 80% comparing treatment and control. We think the standard deviation of the outcome in the study population is 35.

$$
\begin{aligned}
N &= \frac{(Z_{1-\alpha/2} + Z_P)^2 \times V}{\eta^2} \\
&= \frac{(1.96 + 0.84)^2 \times (2 \times 35^2)}{13^2} \\
&= 114
\end{aligned}
$$

Therefore, we need to randomize 114 subjects to each treatment arm of our study (assuming no attrition).

Where do we get the information?

The most difficult part of obtaining the information necessary to undertake sample size calculations is generally getting an estimate of the variance.

This can be done in a few different ways:

- Search the literature
- Use approximate methods to estimate the variance. E.g. for approximately normally distributed variables, the standard deviation is roughly 1/4 of the range of the 95% most common values
- Conduct a pilot study.
- Dichotomize the variable (see below)
- Make an educated guess
- Consider multiple plausible scenarios

Power

What if instead of calculating sample size, we have a fixed sample size N in each group, and we want to calculate power to detect a treatment effect of magnitude η?

We can re-arrange the sample size formula[1] to get

$$P = \Phi\left[\eta\sqrt{\frac{N}{V}} - Z_{1-\alpha/2}\right]$$

- where $Z_{1-\alpha/2}$ is the $(1-\alpha/2)$th quantile of a standard normal variable (e.g. $Z_{1-\alpha/2} = 1.96$ for $\alpha = 0.05$)
- $\Phi(\cdot)$ is the cumulative distribution function for a standard normal random variable
- $V = 2\sigma^2$ where σ^2 is the variance of Y

[1] See Supplemental Slides for the algebra

Power

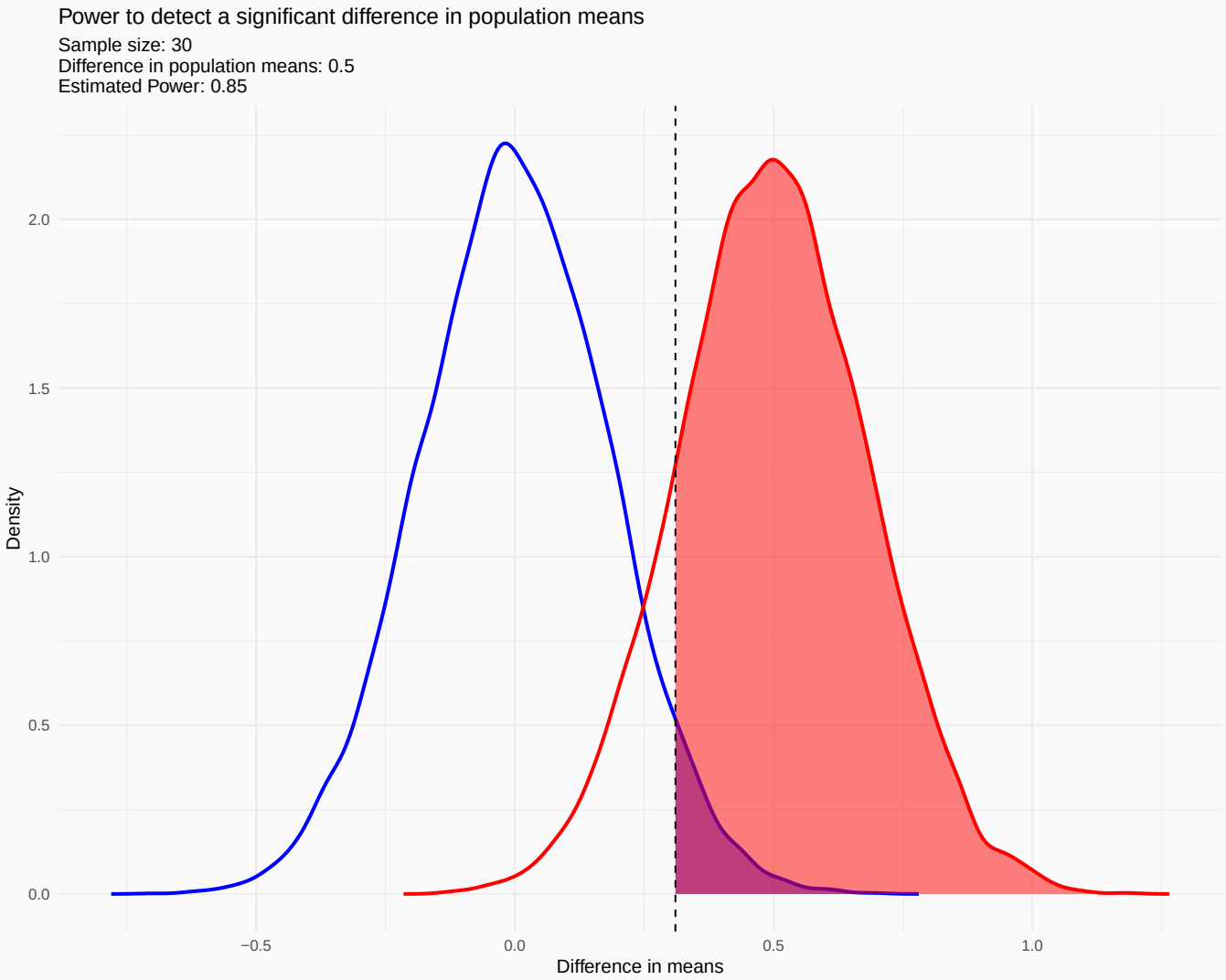
Remembering that $V = 2\sigma^2$ and $\eta = \mathbb{E}[Y \mid A = 1] - \mathbb{E}[Y \mid A = 0] = \mu_1 - \mu_0$, we can also write this as

$$P = \Phi\left[\frac{|\mu_1 - \mu_0|\sqrt{N}}{\sqrt{2}\sigma} - Z_{1-\alpha/2}\right]$$

What happens to power when

- sample size, N, goes up?
- variance of Y, σ^2, goes up?
- effect size, $\eta = |\mu_1 - \mu_0|$, goes up?

Power


Power to detect a significant difference in population means
Sample size: 30
Difference in population means: 0.5
Estimated Power: 0.85
Density
2.0
1.5
1.0
0.5
0.0
−0.5
0.0
0.5
1.0
Difference in means

Sample size and power for a binary outcome

For a **binary** outcome Y, to detect a treatment effect of size $\eta = P[Y = 1 \mid A = 1] - P[Y = 1 \mid A = 0]$, we can use the same formulae for sample size and power but use a different value of V

Sample size:

$$N = \frac{(Z_{1-\alpha/2} + Z_P)^2 \times V}{\eta^2}$$

Power:

$$P = \Phi\left[- Z_{1-\alpha/2} + \eta\sqrt{\frac{N}{V}}\right]$$

- where $V = p_0(1 - p_0) + p_1(1 - p_1)$ and $p_0 = P[Y = 1 \mid A = 0]$, $p_1 = P[Y = 1 \mid A = 1]$ **under the alternative**; OR
- where $V = 2p(1 - p)$ and where p is the overall pooled probability of the event ($p = P[Y = 1]$) **under the null**.

Sample size and power considerations

In some studies, it may be that due to limited resources or limited number of available patients, the sample size is infeasible. In such settings there are several things to consider

- Is the effect size unreasonably small or the assumed variance too large?
- Is it reasonable to increase Type 1 error or decrease power?
- Choose continuous primary outcomes
- Choose outcomes that have smaller variance
- Consider surrogate variables
- Choose binary outcomes that are more common

However, all of these approaches are potentially problematic and it is important to be realistic.

- An underpowered study is a bad investment for everyone.
- Sometimes it is best not to undertake the study or to wait until more resources or patients are available

Sample size and power with covariates

In a randomized trial, control for covariates predictive of the outcome will increase power (why?)

Our sample size and power calculations using the above formula will thus be conservative

Perhaps still best to use power calculations without the covariate control because of

- drop-out
- tendency to be over-optimistic about effect size / variance
- data may be used for other purposes (subgroup analysis)

Tests using linear and logistic regression in RCTs are robust to model mis-specification[2] (Rosenblum and van der Laan, 2009)

Improving efficiency of estimates using baseline covariates while maintaining robustness to misspecification is an active area of research (Tsiatis et al., 2008; Zhang et al., 2008; Lu et al., 2008)

[2] Provided that robust variance estimators are used!

Observational studies and covariates

- In observational studies, covariate control is needed for confounding control.
- Effects on power and sample size can go in either direction
    - If the covariates are predictive of the outcome, power can improve
    - If more predictive of the exposure, power can decline
    - Confounding can alter the effect size in either direction
- It is often difficult to establish this in advance
- Simulations can sometimes help assess how power is affected
- With control for numerous correlated covariates, often power goes down
- In planning we may want to be even more conservative in observational studies
- But often observational studies use secondary data and sample size is fixed

Sample size and power

Some relevant further literature (see next slide for references) for power and sample size calculations include:

Outcome	Test	Reference	Resource type
Binary	χ^2	1	Tables
	Normal	2	Formulae
	Fisher's	3	Tables/Formulae
Continuous	Nonparametric	4	Tables
	F-test	5	Formulae
		6	Tables
		7	Program
		8	Graphs
Survival time	Logrank	9	Formulae
		10	Tables

Sample size and power references

1. Fleiss JL. *Statistical Methods for Rates and Proportions*. New York: Wiley, 1981. pp 38-48.
2. Feigl P. A graphical aid for determining sample size when comparing two independent proportions. *Biometrics*. 1978; 34:111-122.
3. Casagrande JT, Pike MC, Smith PG. The power function of the exact test for comparing two binomial distributions. *Appl Stat*. 1976; 27;176-180.
4. Lehmann EL. *Nonparametrics: Statistical Methods Based on Ranks.* San Francisco: Holden-Day, 1975.
5. George SL, Desu MM. Planning the size and duration of a clinical trial studying time to some critical event. *J Chron Dis*. 1974; 27:15-24.
6. Lesser ML, Cento SJ. Tables of power for the F-test for comparing two exponential survival distributions. *J Chron Dis*. 1981; 34-533-544.
7. Bernstein D, Lagakos SW. Sample Size and power determination for stratified clinical trials. *J Stat Comput Simul*. 1978; 8:65-73.
8. Schoenfield DA, Richter JR. Monograms for calculating the number of patients needed for a clinical trial with survival as an endpoint. *Biometrics*. 1982; 38:163-170.
9. Peto R, Pike MC, Armitage P, et al. Design and analysis of clinical trials requiring prolonged observation of each patient. *Br J Cancer*. Part I – 1976; 34:585-612. Part II – 1977; 35:1-39.
10. Freedman LS. Tables of the number of patients required in clinical trials using the logrank test. *Stat in Med*. 1982; 1:121-129.

Sample size and power references

UCLA `http://calculators.stat.ucla.edu/powercalc/`

- Normal, Exponential, Binomial, Poisson
- Sample Size and Power Calculations

MGH: `http://hedwig.mgh.harvard.edu/sample_size/size.html`

- Binomial (Parallel or Cross-over Trial)
- Normal (Parallel or Cross-over Trial)
- Time to Event

Statpages: `http://statpages.org/`

- An excellent compendium of online statistical tools

Cluster Randomized Trials

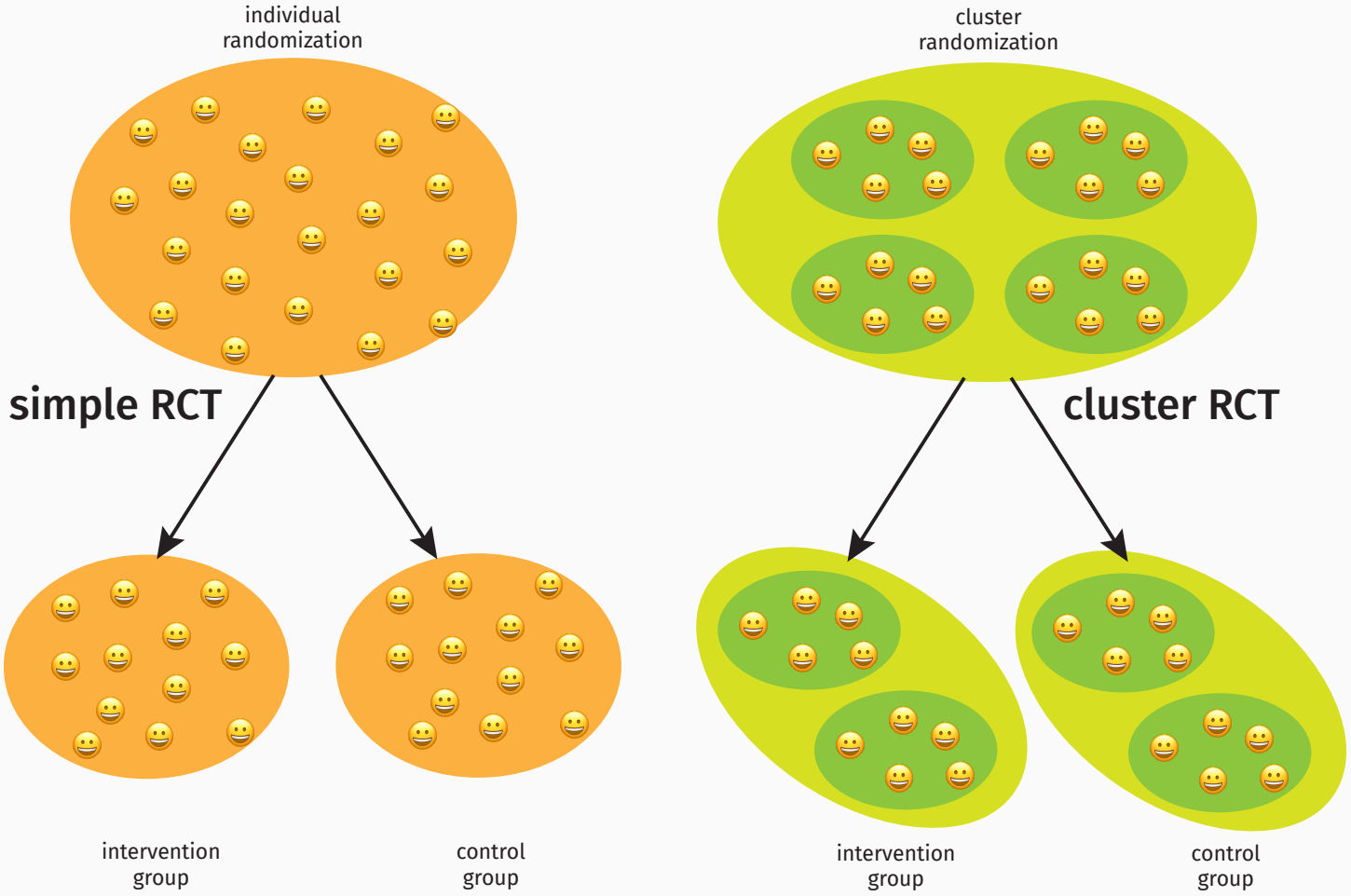
Cluster Randomized Trials

- Individuals are clustered in groups
- The entire group is randomized to treatment or control
- Preserves the advantages of randomization
- But requires special care in design and analysis

Reasons for a Cluster Randomized Trial:

- Administrative convenience
- To avoid treatment group contamination
- Intervention is naturally applied at the cluster level

Cluster randomized trial



Why cluster randomize?

- **Logistic convenience and acceptability:** the intervention by its nature has to be applied to entire communities or other groupings of individuals, or it is more convenient or acceptable to apply it in this way.
  - Example 1: in a rural area of Bangladesh, a study is conducted to look at the effects of enhanced micro-credit services on utilization of health services. Although the lending process is carried out at an individual level, it would not seem fair for one woman to have access to credit while her neighbor does not.
  - Example 2: A new program of peer-led sex education in secondary schools involves training older students to act as peer-educators and to work with classes of younter students during scheduled classroom sessions. It would not be possible to randomize individuals to receive this intervention, as it is designed for delivery to entire classes.

Why cluster randomize?

- We wish to avoid contamination that might result if individuals in the same community were to be randomized to different treatment arms.
- **Contamination:** refers to the situation where individuals in one treatment arm actually receive part or all of the intervention allocated to another treatment arm.
  - Example: Secondary school students who are smokers are randomly allocated to receive individual counseling using cognitive behavioral therapy (CBT) to help them give up smoking, or to a control group. Students in the CBT group may talk to friends in the control group about the counseling they have received.

Why cluster randomize?

- We wish to capture the **indirect or spillover effects** of intervention
  - Example: The Primary School Deworming Project was an intervention consisting of deworming treatments for intestinal helminths such as hookworm, roundworm, whipworm, and schistosomiasis, delivered to school children in an area of Kenya with high helminth infection rates. The primary goal was to to evaluate whether the deworming treatments resulted in reduced instestinal helminth infections among students. The intervention was administered at the school level and all students in treated schools were offered medication.
  - The medication was believed to have both direct and spillover effects: it not only killed helminths among those who took them (i.e. direct effect), but also decreased disease transmission among peers by reducing the number of helminths in the environment (i.e. a spillover effect).

Unit of Randomization vs. Unit of Analysis

- A key property of **cluster randomized trials** is that inferences are usually intended to apply at the individual level while randomization is at the cluster or group level. Thus, the unit of randomization may be different from the unit of analysis.
- In this case, the **lack of independence** among individuals in the same cluster creates special methodological challenges in both design and analysis.
- Another key feature of cluster randomized trials is that, because clusters are often large and there may not be many of them, it is often not possible to randomize a large number of clusters to each treatment arm. With small numbers of clusters, we cannot rely on randomization to ensure adequate balance between arms.

Implications of Clustering

- Reduction in effective sample size
  - the extent depends on the degree of within-cluster correlation and on average cluster size
- Standard approaches for sample size estimation and statistical analysis do not apply
  - application of standard sample size approaches leads to an underpowered study
  - random effects models or generalized estimating equations (GEE) are often used to deal with clustering in analysis
  - application of standard statistical methods generally tends to underestimate standard errors, i.e. bias p-values downwards

Implications of Clustering

- Consider a trial in which c clusters of size m are randomly assigned to each of an experimental and control arm. Also assume that the response variable Y is normally distributed with common variance σ^2.
- The aim is to test $H_0 : \mu_1 = \mu_0$. Then appropriate estimates of μ_1 and μ_0 are given by $\bar{y}_1$ and $\bar{y}_0$, the usual sample means.
- We would usually test this hypothesis with a t-test
- How should we think about the implications of clustering?

Design Effect and Effective Sample Size

Recall from our discussion of **cluster sampling** in PHS2000A:

- Cluster sampling provides less precise estimates of parameters of interest than simple random sampling

$$\begin{aligned}\text{Design Effect} &= \frac{\text{Variance under cluster sampling}}{\text{Variance under simple random sampling}} \\ &= 1 + (m-1)\rho\end{aligned}$$

  where m is the cluster size and ρ is the intracluster correlation coefficient.

- Design effect ($DEFF$) increases with ρ and m; we can think of it as a **variance inflation factor**
- If c clusters of size m are randomized to each of two treatment arms,

$$\text{Effective sample size} = \frac{cm}{DEFF}$$

- What would happen to the power of the study if we use standard sample size calculation formulae, ignoring the DEFF?

Intracluster correlation coefficient ρ

The overall response variance σ^2 may be expressed as the sum of two components, i.e.

$$\sigma^2 = \sigma_B^2 + \sigma_W^2$$

where

$$\sigma_B^2 = \text{between-cluster component of variance}$$
$$\sigma_W^2 = \text{within-cluster component variance}$$

then

$$\rho = \frac{\sigma_B^2}{(\sigma_B^2 + \sigma_W^2)}$$

Comparison of means

Suppose c clusters of size m are to be assigned to each of two intervention groups. Then the number of subjects required per intervention group at significance level α and power β to test $H_0 : \mu_1 = \mu_0$ is given by

$$N = cm = \frac{(Z_{1-\alpha/2} + Z_P)^2 * V}{(\mu_1 - \mu_0)^2}$$

where

- $V = (2\sigma^2)[1 + (m-1)\rho]$
- $\sigma^2 = \sigma_B^2 + \sigma_W^2$
- $\rho = \sigma^2/(\sigma_B^2 + \sigma_W^2)$

The required number of clusters per intervention group is given by $c = N/m$.

Estimating ρ

During the planning process, if data are available from the study clusters, or else from comparable units in a different but similar population, these data can be used to obtain estimates of σ_B^2 and σ_W^2. For example, for a continuous outcome Y, we could fit a **linear mixed effects model**:

$$
\begin{aligned}
Y_{ij} &= \beta_0 + u_{0j} + \epsilon_{ij} \\
u_{0j} &\sim N(0, \sigma_{u_0}^2) \\
\epsilon_{ij} &\sim N(0, \sigma_\epsilon^2) \\
u_{0j} &\perp\!\!\!\perp \epsilon_{ij}
\end{aligned}
$$

(using the notation from our discussion of multilevel models in PHS2000A).

Note that the **between-cluster variance** is $\sigma_B^2 = \sigma_{u_0}^2$ and the **within-cluster variance** is $\sigma_W^2 = \sigma_\epsilon^2$.

Example

Hsieh (1988) reported on the results of a pilot study for a planned 5-year trial examining cardiovascular risk factors, obtaining cholesterol levels from 754 individuals in 4 worksites.

- Estimated variance components were

$$\hat{\sigma}_W^2 = 2209 \quad \text{and} \quad \hat{\sigma}_B^2 = 93$$

- Therefore the value of ρ was assessed as

$$\rho = \frac{93}{(93 + 2209)} = 0.04$$

- Assuming $m = 70$ subjects/worksite,

$$DEFF = 1 + (70 - 1)0.04 = 3.76$$

Example

- To obtain 80% power at $\alpha = 0.05$ (2 sided) for detecting a mean difference of 20 mg/dL between intervention and control groups, the number of required worksites per treatment arm is given by

$$
\begin{aligned}
N &= \frac{(Z_{1-\alpha/2} + Z_P)^2 \times (2\sigma^2) \times [1 + (m-1)\rho]}{(\mu_1 - \mu_0)^2} \\
&= \frac{(1.96 + 0.84)^2 \times 2(2302) \times (3.76)}{20^2} \\
&= 339
\end{aligned}
$$

- If $N = cm$, then the number of clusters required is

$$c = \frac{N}{m} = \frac{339}{70} = 4.8$$

So ~ 5 clusters per treatment arm would be needed (one might want to increase this number to account for possible loss to followup).

Number of clusters and Number of subjects per cluster

Number of clusters (c)	Subjects/cluster (m)	Total subjects ($N = cm$)	DEFF[3]	Effective sample size	Power[4]
4	320	1280	6.42	199	0.514
8	160	1280	3.70	346	0.749
16	80	1280	2.34	547	0.911
32	40	1280	1.66	771	0.975
64	20	1280	1.32	970	0.993
128	10	1280	1.15	1113	0.997
1280	1	1280	1	1280	0.999

[3] assuming $\rho = 0.017$

[4] assuming $\eta = 0.2$

Practical Considerations

Trial randomizing between 30 and 50 individuals per cluster will tend to have almost the same statistical power as trials randomizing the same number of much larger units

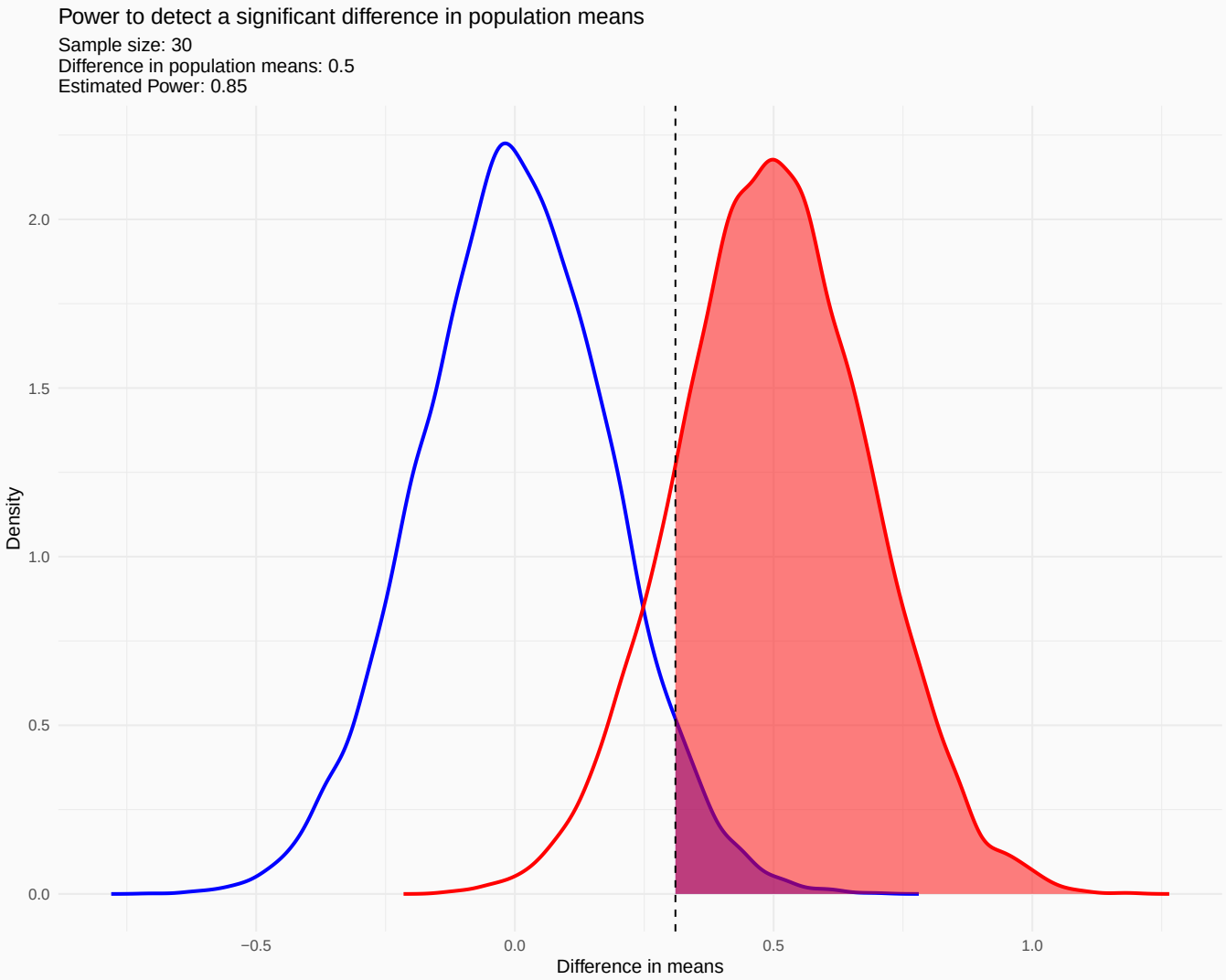
- But clusters of larger size are often recruited for practical reasons

Another approach to cluster randomized trials is to pair match based on available covariate and then randomize

- This has the advantage of generally making treated and control groups more comparable
- When sample sizes of clusters dramatically differ this can be especially helpful and sample size of the cluster can be taken as a matching variable (Imai et al., 2009)

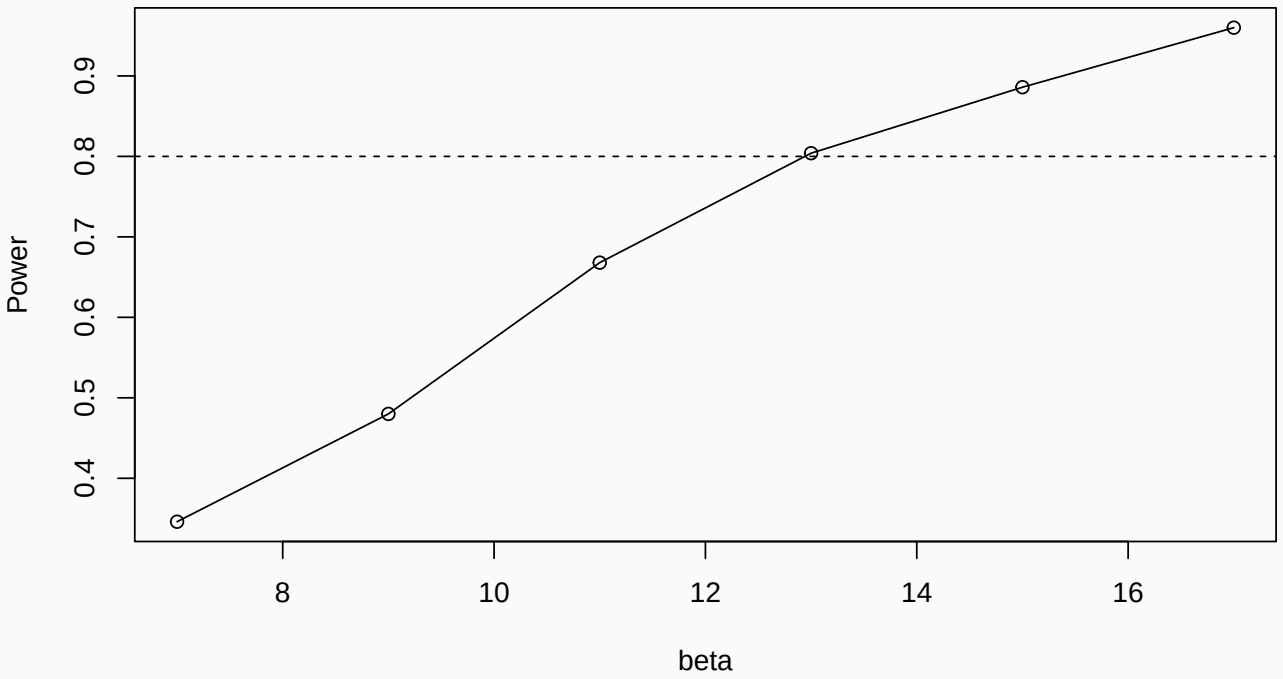
A simulation approach

Let's consider the LDL cholesterol example from the beginning of class. Given a sample size of $n = 114$ per arm, we want to explore power to detect a range of possible effects of the cholesterol lowering drug, from a 7 point decrease to a 17 point decrease, given a standard deviation of 35.



```r
library(tidyverse)
library(broom)

n_sim <- 500
n_per_arm <- 114
beta <- seq(7, 17, by=2)

results <- array(NA, dim = c(n_sim, length(beta)))

df <- data.frame(trt = c(rep(0, n_per_arm), rep(1, n_per_arm)))


for (j in 1:length(beta)) {
  for (i in 1:n_sim) {
    # generate data
    df$y <- 140 - beta[j]*df$trt + rnorm(2*n_per_arm, 0, 35)

    # fit a linear regression model
    this_model <- lm(y ~ trt, data=df)

    results[i, j] <- broom::tidy(this_model)$p.value[2]
  }
}

apply(results<0.05, 2, sum)/n_sim
[1] 0.346 0.480 0.668 0.804 0.886 0.960

plot(beta, apply(results<0.05, 2, sum)/n_sim, ylab = "Power")
lines(beta,apply(results<0.05, 2, sum)/n_sim)
abline(h=0.8, lty = 2)
```

A simulation approach



Supplemental Slides

Rearranging the sample size formula to get power

If we want to compute power instead of sample size, we can re-arrange the expression. Remembering that $V = 2\sigma^2$ and $\eta = \mu_1 - \mu_0$

$$
\begin{aligned}
N &= \frac{(Z_{1-\alpha/2} + Z_P)^2 \times V}{\eta^2} \\
N(\mu_1 - \mu_0)^2 &= (Z_{1-\alpha/2} + Z_P)^2 \times 2\sigma^2 \\
\frac{N(\mu_1 - \mu_0)^2}{2\sigma^2} &= (Z_{1-\alpha/2} + Z_P)^2 \\
Z_{1-\alpha/2} + Z_P &= \sqrt{\frac{N(\mu_1 - \mu_0)^2}{2\sigma^2}} \\
Z_P &= \frac{|\mu_1 - \mu_0|\sqrt{N}}{\sqrt{2}\sigma} - Z_{1-\alpha/2} \\
P = 1 - \beta &= \Phi\left(\frac{|\mu_1 - \mu_0|\sqrt{N}}{\sqrt{2}\sigma} - Z_{1-\alpha/2}\right)
\end{aligned}
$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function.

Sample size and power for interactions

Sample size for interactions in logistic regression models

Consider the logistic regression model:

$$logit\,[P(Y = 1 \mid G = g, E = e)] = \beta_0 + \beta_1 g + \beta_2 e + \beta_3 g * e$$

Suppose we were interested in the sample size required to detect a log interaction odds ratio of $\beta_3 = \eta$

We can do so by specifying:

- The significance level α (e.g. 0.05) and power P (e.g. 0.80)
- The main effect log odds ratios β_1 and β_2
- The proportions in each $G \times E$ exposure strata $\pi_{ij} = P[G = i, E = j]$

Instead of the proportions π_{ij} we can alternative specify

- the overall prevalence of G and E, i.e $p_g = P(G = 1)$, $p_e = P(E = 1)$,
- the odds ratio relating G and E,

$$OR_{ge} = \frac{P(G = 1|E = 1)/P(G = 0|E = 1)}{P(G = 1|E = 0)/P(G = 0|E = 0)}$$

  and
- $P(Y = 1|G = 0, E = 0)$ (see Appendix slides)

This latter specification will be easier to use in a case-control study.

Note that $\beta_0 = logit[P(Y = 1|G = 0, E = 0)]$ where $P(Y = 1|G = 0, E = 0)$ is the prevalence of cases in the doubly unexposed group in the study (e.g. in the actual case-control sample)

Sample size for interactions in logistic regression models

To calculate the overall sample size (cases and controls) required to detect a log interaction odds ratio of $\beta_3 = \eta$ with significance level α and power P, we can use

$$N = \frac{(Z_{1-\alpha/2} + Z_P)^2 \times V}{\eta^2}$$

where $Z_{1-\alpha/2}$ and Z_P denote the $(1-\alpha/2)$th and Pth quantiles of a standard normal variable, e.g. $Z_{1-\alpha/2} = 1.96$ for $\alpha = 0.05$ and $_P = 0.84$ for $P = 0.80$.

And where V is the variance of the estimator of β_3 evaluated under the alternative hypothesis (see next slide).

Sample size for interactions in logistic regression models

The variance V takes the form (Demidenko, 2008)

$$V = \frac{1}{L} + \frac{1}{R} + \frac{1}{F} + \frac{1}{J}$$

where

$$L = \frac{exp(\beta_0)}{[1 + exp(\beta_0)]^2} \pi_{00}$$

$$F = \frac{exp(\beta_0 + \beta_1)}{[1 + exp(\beta_0 + \beta_1)]^2} \pi_{10}$$

$$J = \frac{exp(\beta_0 + \beta_2)}{[1 + exp(\beta_0 + \beta_2)]^2} \pi_{01}$$

$$R = \frac{exp(\beta_0 + \beta_1 + \beta_2 + \eta)}{[1 + exp(\beta_0 + \beta_1 + \beta_2 + \eta)]^2} \pi_{11}$$

Power for interactions in logistic regression models

If instead of sample size we want to calculate power to detect a log interaction odds ratio of magnitude η from a given dataset with sample size N, the following formula can be used

$$P = \Phi\left[\eta\sqrt{\frac{N}{V}} - Z_{1-\alpha/2}\right]$$

- where $Z_{1-\alpha/2}$ is the $(1-\alpha/2)$th quantile of a standard normal variable (e.g. $Z_{1-\alpha/2} = 1.96$ for $\alpha = 0.05$)
- $\Phi(\cdot)$ is the cumulative distribution function for a standard normal random variable
- V is obtained by the expressions given above

Sample size and power for interactions in logistic regression models

- These can all be done by hand using the formulae on the previous slides.
- Alternatively there is software available for this at: `http://www.dartmouth.edu/~eugened`
- Demidenko's spreadsheet is relatively straightforward for cohort studies
- However, for case-control data it requires specifying $P(Y = 1|G = 0, E = 0)$ i.e. the prevalence of cases in the doubly unexposed group in the case-control sample
- This is not a particularly straightforward quantity to specify
- We could obtain it by specifying the proportion of cases to control, along with $p_g = P(G = 1)$, $p_e = P(E = 1)$, and the odds ratio relating G and E, but this then requires solving a 'quartic' equation (Demidenko, 2008)
- Alternatively, under a rare outcome assumption, we can obtain simple analytic expression (VanderWeele, 2012). There is an Excel spreadsheet (available on Canvas) to implement this automatically

Sample size and power for interactions in logistic regression models

- Prior to Demidenko (2008), there had been a number of previous sample size calculations for interactions in logistic regression (Hwang et al., 1994; Foppa and Spiegelman, 1997)
- However, these calculated the variance under the null rather than the alternative (Garcia-Closas and Lubin, 1999; Demidenko, 2008) which led to underestimates of the sample size when the interaction ratio was relatively large if variance is calculated under the alternative (as it is in nearly all software!)
- The sample size calculation above are for Wald-tests (standard SAS/Stata/R output) with variance under the alternative
- Lubin and Gail (1990) describe a method for the score test with variance under the alternative
- Gauderman et al. (2002ab) developed sample size calculations for various matched designs and family-based designs us likelihood ratio tests

Sample size for additive interaction

Consider again the logistic regression model:

$$logit\,[P(Y = 1|G = g, E = e)] = \gamma_0 + \gamma_1 g + \gamma_2 e + \gamma_3 g * e$$

Suppose we were interested in the sample size required to detect **additive interaction** of magnitude $RERI = \eta$

We can again do so by specifying:

- The significance level α (e.g 0.05) and power P (e.g. 0.80)
- The main effect log odds ratios γ_1 and γ_2
- The proportions in each $G \times E$ exposure strata $\pi_{ij} = P(G = i, E = j)$

Or, instead of (3) by specifying:

- the overall prevalence of G and E i.e $p_g = P(G = 1)$, $p_e = P(E = 1)$,
- the odds ratio relating G and E
- The proportion of cases to controls in the study

Sample size for additive interaction

To calculate the overall sample size required to detect additive interaction of $RERI = \eta$ with significance level α and power P we can use:

$$N = \frac{(Z_{1-\alpha/2} + Z_P)^2 \times V}{\eta^2}$$

where $Z_{1-\alpha/2}$ and Z_P denote the $(1-\alpha/2)$th and Pth quantiles of a standard normal variable, e.g. $Z_{1-\alpha/2} = 1.96$ for $\alpha = 0.05$ and $_P = 0.84$ for $P = 0.80$.

And where V is the variance of the estimator $RERI$ given by (VanderWeele, 2012):

$$V_{RERI(OR)} = \left[\frac{1}{L} + \frac{1}{R}\right] e^{2(\gamma_1+\gamma_2+\gamma_3)} - \frac{2}{L} e^{2\gamma_1+\gamma_2+\gamma_3} - \frac{2}{L} e^{\gamma_1+2\gamma_2+\gamma+3} + \left[\frac{1}{L} + \frac{1}{F}\right] e^{2\gamma_1} + \left[\frac{1}{L} + \frac{1}{J}\right] e^{2\gamma_2} + \frac{2}{L} e^{\gamma_1+\gamma_2}$$

with L, R, F, J as given above and in Demidenko (2008)

As before power for a given sample size N is obtained by:

$$P = \Phi\left[\eta\sqrt{\frac{N}{V}} - Z_{1-\alpha/2}\right]$$

Sample size for additive interaction

Similar expressions (VanderWeele, 2012) are available for sample size or power if

1. RERI is calculated from a log-linear model rather than a logistic model
2. RERI is calculated using logistic regression with case-control data assuming a rare outcome
3. Additive interaction using risks are used

Slightly different expressions are needed to estimate π_{ij} which are needed to calculate L, R, F, J which are needed to obtain the variance V for RERI (VanderWeele, 2012)

Sample size and power for interactions

- Excel spreadsheets are available (VanderWeele, 2012) to calculate power and sample size for additive interaction for either cohort studies (using risks or RERI) or for case control studies (using RERI) or for multiplicative interaction and for case-only estimators of interaction and these comparisons can be informative (see Supplemental slides)
- Power is often very low for interaction analyses
- There will likely be many false positives (as has played out in $G \times E$ literature)
- Rule of Thumb: 4 Times the Sample Size is required to detect interaction
- This has implications for when one ought to undertake an interaction analysis
- Often it will be inappropriate if a study is only powered to detect a main effect
- One needs caution with regard to RCT subgroup analyses, especially if these are not specified *a priori*
- In general whenever main effects and the interaction is positive, there will be more power to detect additive interaction than multiplicative interaction,
    - especially when the interaction is modest but the main effects are large
    - when outcome probabilities are additive or sub-additive power for multiplicative interaction is sometimes greater (Greenland, 1983)

Sample size for interactions in logistic models

If instead of the proportions $\pi_{ij} = P(G = i, E = j)$ we specify

- the overall prevalence of G and E, i.e. $p_g = P(G = 1)$ and $p_e = P(E = 1)$,
- the odds ratio relating G and E,

$$\Delta = \frac{P(G = 1|E = 1)/P(G = 0|E = 1)}{P(G = 1|E = 0)/P(G = 0|E = 0)}$$

- and $P(Y = 1|G = 0, E = 0)$

we can calculate π_{ij} as follows:

$$\pi_{00} = \frac{(1 - p_e)}{(1 + C)}$$

$$\pi_{01} = \frac{p_e}{(1 + C\Delta)}$$

$$\pi_{10} = \frac{C(1 - p_e)}{(1 + C)}$$

$$\pi_{11} = \frac{p_e C\Delta}{(1 + C\Delta)}$$

where

$$C = \frac{q + \sqrt{q^2 + 4p_g(1 - p_g)\Delta}}{2(1 - p_g)\Delta}$$

and

$$q = p_g(1 + \Delta) + p_e(1 - \Delta) - 1$$

Sample size for case-only interactions

Now consider a log-linear model

$$log\,[Pr(D = 1|G = g.E = e)] = \beta_0 + \beta_1 g + \beta_2 e + \beta_3 ge$$

and the case-only estimator of β_3 under $G \times E$ independence:

$$\beta_3 = \frac{P(G = 1|E = 1, D = 1)/P(G = 0|E = 1, D = 1)}{P(G = 1|E = 0, D = 1)/P(G = 0|E = 0, D = 1)}$$

Suppose we were interested in the sample size required to detect a log interaction ratio of $\beta_3 = \eta$

We can do so by specifying:

- The significance level α (e.g. 0.05) and power P (e.g. 0.80)
- The main effect risk ratios for the genetic factor alone $R_g = exp(\beta_1)$ and for the environmental factor alone $R_e = exp(\beta_2)$
- The overall proportions of G and E in the population; let $g = P(G = 1)$ and $e = P(E = 1)$

Sample size for case-only interactions

To calculate the sample size required to detect a log interaction ratio of $\beta_3 = \eta$ with significance level α and power P we can once again use:

$$N = \frac{(Z_{1-\alpha/2} + Z_P)^2 \times V}{\eta^2}$$

where $Z_{1-\alpha/2}$ and Z_P denote the $(1-\alpha/2)$th and Pth quantiles of a standard normal variable, e.g. $Z_{1-\alpha/2} = 1.96$ for $\alpha = 0.05$ and $_P = 0.84$ for $P = 0.80$,

and where V is the variance of the case-only estimator of β_3 evaluated under the alternative.

Yang et al. (1999) give a sample size formula for the case-only estimator but the variance is given under the null rather than the alternative. On the following slide we will instead give the variance under the alternative.

Sample size for case-only interactions

$$N = \frac{(Z_{1-\alpha/2} + Z_P)^2 \times V}{\eta^2}$$

For the case-only estimator, V under the alternative is given by (VanderWeele, 2011; cf. Yang et al., 1999):

$$v_A = (m_1 + m_2 + m_3 + m_4)\left(\frac{1}{m_1} + \frac{1}{m_2} + \frac{1}{m_3} + \frac{1}{m_4}\right)$$

where

$$\begin{aligned} m_1 &= (1-g)(1-e) \\ m_2 &= g(1-e)R_g \\ m_3 &= (1-g)eR_e \\ m_4 &= geR_gR_eR_i \end{aligned}$$

Sample size for case-only interactions

Yang et al. (1997) give a sample size formula for the case-only estimator but the variance is given under the null rather than the alternative and it will again underestimate the sample size for large interaction ratios. Nonetheless their comparison with sample sizes for logistic regression and case-control studies is informative.

Prevalence of genotype		$R_i^* = 2$ Exposure			
		0.1	0.3	0.5	0.7
0.05	$N_{c\text{-}c}$	4,282	1,929	1,708	2,147
	$N_{c\text{-}o}$	2,996	1,223	997	1,167
0.10	$N_{c\text{-}c}$	2,293	1,041	926	1,167
	$N_{c\text{-}o}$	1,568	659	552	662
0.30	$N_{c\text{-}c}$		486	442	569
	$N_{c\text{-}o}$		308	284	372
0.50	$N_{c\text{-}c}$			414	549
	$N_{c\text{-}o}$			284	400
0.70	$N_{c\text{-}c}$				757
	$N_{c\text{-}o}$				600

Table shows the number of cases required for case-control ($N_{C\text{-}C}$) and case-only ($N_{C\text{-}O}$) studies to detect gene-environment interaction for $\alpha = 0.05$, Power = 0.80, $R_g = R_e = 1$, $exp(\eta) = 2$.

Sample size and power for mediation

Sample size for mediation

- Very little formal work on sample size calculations for direct and indirect effects
- Many papers use and cite simulation results from: Fritz MS, MacKinnon DP. (2007) Required Sample Size to Detect the Mediated Effect. *Psychological Science* 18:193-198.
- Helpful Rule of Thumb (Kenny and Judd, 2014): In many (but not all!) contexts, power to detect indirect effect is greater than the total effect; but power to detect the direct effect is less than the total effect
    - Can be helpful in planning mediation analyses
    - Also, because of power, one must be careful of claims of "no direct effect" and therefore "complete mediation"
- There has been some recent work and tools on simulation-based approaches for mediation power and sample size calculations (Rudolph et al., AJE 2020)